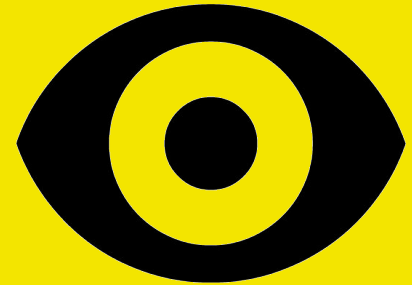


The AI Brief

An overview of the UK's
Artificial Intelligence sector



Beauhurst

This report analyses the ambitious private companies operating in the UK's artificial intelligence sector and the supporting mechanisms for growth. With almost 700 active ambitious companies, and a healthy supply of capital from investors across the ecosystem, AI is becoming an ever increasing strength of the UK high-growth economy.

Artificial Intelligence, or AI, encompasses a range of computer programmes intended to solve problems requiring learning, understanding context or otherwise operating with autonomy from humans using human-like thought processes and the processing power of computers.

As we'll see in this overview of the sector, AI has been around for a while, but has only really taken off within the last five years. Technology has developed drastically in this time, and with huge investor appetite the private market is truly flourishing. As discussion increasingly turn towards the dangers and responsibilities that come with powerful AI technology, governments are also supporting and preparing to shape growth in the industry.

There are a couple of reasons for this; to stay ahead of the international competition, in terms of both national security and establishing the country as a global thought leader to attract further investment; and to form appropriate legislation that protects citizens, whilst nurturing an environment conducive to innovation. As of yet, regulations remain lax, allowing innovators to explore the full range of opportunities that ambitious AI companies can offer.

What do we mean by “ambitious”?

We use 8 criteria to identify high-growth and ambitious companies. Each of these is a sign of external validation that a business is either actively growing or creating ambitious growth plans.

All companies featured in this report have met at least one of these growth triggers.



Equity investment



Venture debt



Accelerator attendance



Grant recipients



Scaleups



High-growth lists



MBO/MBI



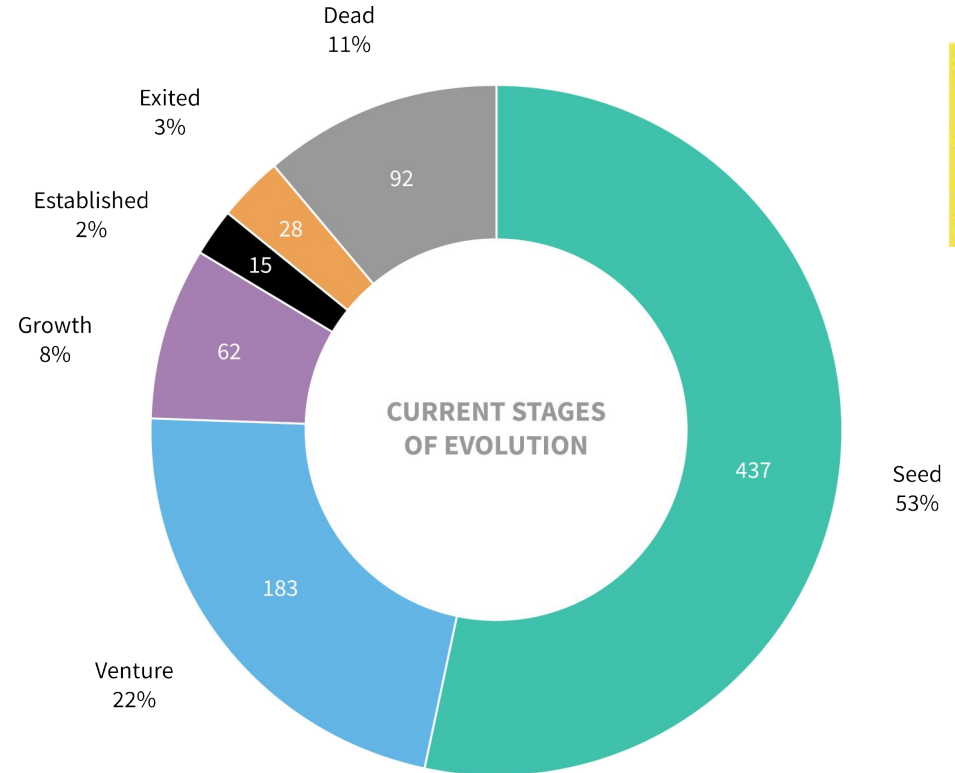
Academic spinouts

Stages of evolution

There are **697** active, ambitious AI companies in the UK.

An additional **28** have exited the market, either through an IPO or an acquisition, and **92** are no longer trading.

With over **50%** of AI companies in the seed stage, and few reaching the later stages of growth at this point, it's clear that the sector is nascent. Each of these companies has the potential to transform society with their proprietary tech, but most will need to use external support to reach commercialisation.



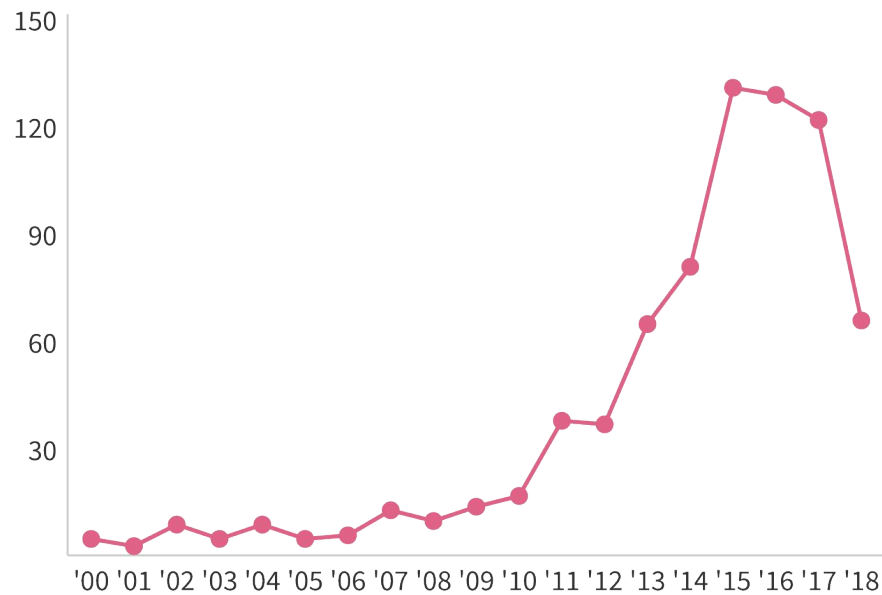
Age of the sector

Indeed, the sector has seen a sudden surge in popularity over the past few years, with **130** high-growth AI companies having been incorporated in 2015 alone.

The decline in number since then is simply a time lag; companies incorporated in the last three years have not had as much time to gain traction and exhibit high-growth.

The race is on for investors to pick out the most promising early stage ventures from the rapidly growing crowd.

NUMBER OF HIGH-GROWTH AI COMPANIES INCORPORATED



Methods of growth

AI is an IP driven sector; **9%** of high-growth AI companies have spun out of an academic institution, and **30%** have received innovation grants. These figures are significantly higher than the sector agnostic average of 3% and 17% respectively.

These companies are backed by notable amounts of equity, with **76%** raising investment since 2011, compared to just 53% across all sectors.

At **39%**, AI also has a high rate of accelerator attendances (compared to 22% across all high-growth companies).

Raised equity

622

Visible scaleups

2

Received grants

247

Spun out of a university

71

Attended accelerators

319

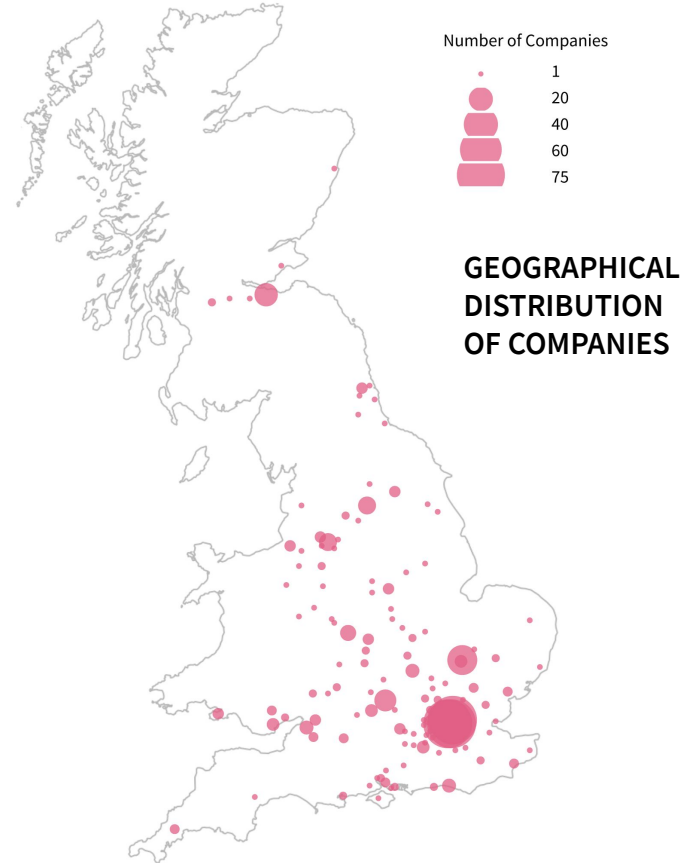
Featured on a high-growth list

64

Geographical distribution

While London dominates the UK's AI scene with **505** companies (62%) headquartered in the capital, there are also substantial hubs around Cambridge (**28**), Edinburgh (**17**) and Oxford (**15**).

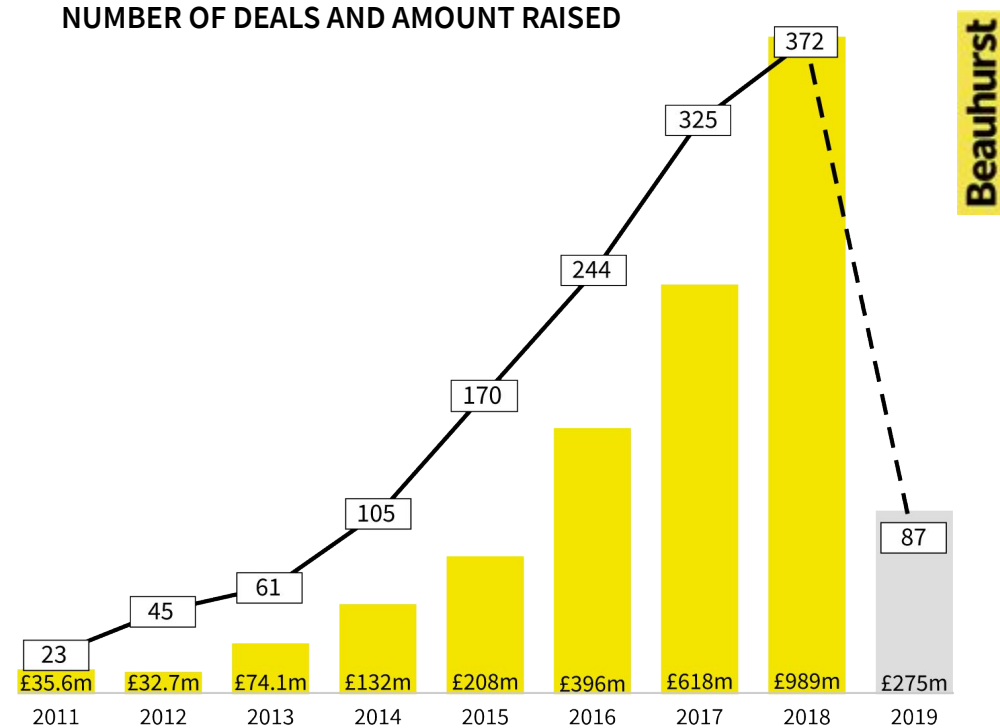
It's no coincidence that these cities all host world class universities. 12 of the of AI startups in Oxford (80%) have been spun out from academic institutions. There are only five in Cambridge, but a further six companies have spun out from the University of Cambridge and chosen headquarters in other cities, including leading AI unicorn, Darktrace.



Fundraising statistics

AI companies have secured consistently increasing amounts of equity investment since 2011. It is one of the few sectors to have raised more money in 2018 than 2017 *and* secure a higher volume of deals.

Using Q1 2019 investment figures as a reference, we expect funding levels to surpass the £1b mark for the first time. With just **87** deals completed in the first quarter of this year, the number of deals completed by the end of the year is less likely to match the record level in 2018.

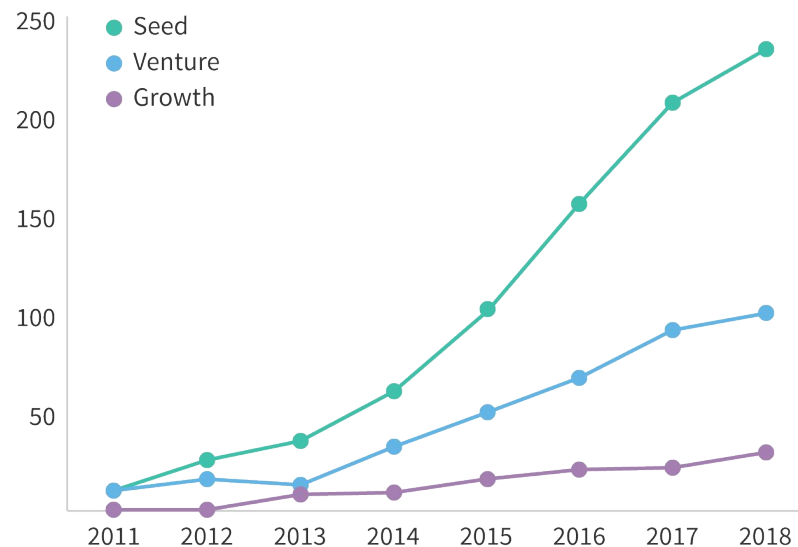


Fundraising statistics

The number of deals secured by seed-stage companies was down across all sectors in 2018 – even in fintech. But AI managed to buck this trend, securing **13%** more seed stage deals in 2018 than in 2017.

This is incredibly reassuring to see amidst such economic uncertainty, and shows there is substantial support for ambitious AI companies choosing to grow in the UK. Seed funding is especially important for a sector with such a focus on research and development, helping companies to survive beyond this capital intensive phase.

NUMBER OF DEALS BY STAGE OF EVOLUTION

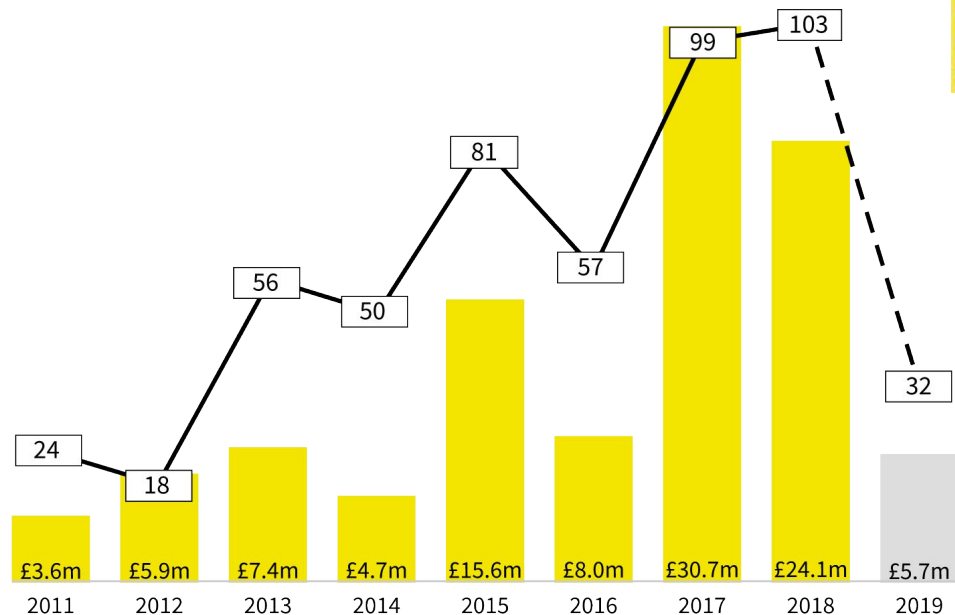


Grant statistics

In an effort to establish the UK as a global leader in AI, the government is seeking ways to nurture ambitious companies in the sector. These plans were set out in the [Industrial Strategy](#), published in 2017.

Reflecting these intentions is the growing number of grants allocated to AI companies by government bodies such as Innovate UK. Considering the majority of grants tend to be allocated later in the year, 2019 is on track to see more grants allocated than ever before, in terms of both number and value.

NUMBER OF GRANTS AND AMOUNT ALLOCATED



Top investors

Parkwalk Advisors are the most prolific investors into UK AI since 2011, with **19** deals across **8** companies. Their portfolio includes University of Oxford spinout Oxbotica, insurtech platform Cytora, and adtech company Mirriad, which IPO'd on the AIM for £25.4m in December 2017.

Meanwhile, Amadeus Capital Partners have the highest number of AI companies in their portfolio, with **17** announced funding rounds across **11** companies, including phishing software Tessian and unicorn startup Graphcore.

TOP FUND MANAGERS BY NUMBER OF DEALS SINCE 2011

Parkwalk Advisors	19
Amadeus Capital Partners	17
Mercia Technologies	15
LocalGlobe	11
IQ Capital Partners	11

Top companies by equity received and pre-money valuation

GRAPHCORE

Graphcore has developed a processor optimised for machine-learning tasks.

Location

Bristol City Council

Stage of Evolution

Growth

Total amount raised

£244m

Latest pre-money valuation

£925m



Darktrace has developed security software that identifies anomalies that could signal security threats.

Location

Westminster City Council

Stage of Evolution

Growth

Total amount raised

£173m

Latest pre-money valuation

£1.47b

BenevolentAI

BenevolentAI aims to speed up scientific discovery through mass data analysis of data with AI.

Location

Camden London Borough Council

Stage of Evolution

Venture

Total amount raised

£147m

Latest pre-money valuation

£1.49b

Top companies by value of grants received



FiveAI is developing artificial intelligence software to be used in autonomous driving technology.

Location

Bristol City Council

Stage of Evolution

Venture

Total Value of Grants Received

£9.3m

Number of Grants Received

1



Oxbotica develops software and technology to power driverless vehicles.

Location

Oxford City Council

Stage of Evolution

Venture

Total Value of Grants Received

£7.4m

Number of Grants Received

10



Vivacity Labs uses artificial intelligence to develop traffic monitoring software.

Location

Camden London Borough Council

Stage of Evolution

Venture

Total Value of Grants Received

£5.91m

Number of Grants Received

7

Access all this data with a subscription to Beauhurst

Beauhurst is a searchable database of the UK's most ambitious companies – from seed-stage startups to growing scaleups and beyond.

We collect and manually curate hundreds of data-points on over 23,000 companies. Our new Explore page allows you to access all this information, live.

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